

Tribhuvan University

Saraswati Multiple Campus
Lekhnath marg, Thamel

Pre-Board Examination (2082)

Group-B

Attempt any six questions (6X5=30)

1. What is JavaScript? Write a JavaScript code to print smallest and largest number among 10 elements of an array?
2. Explain the two HTTP functions to accept the user values from interface with suitable PHP program?
3. What is CMS? Write a steps to create submenu in WordPress or Joomla?
4. What is Ajax? Write down advantage and disadvantages of Ajax?
5. What is super global variable? Describe the various super global variable in PHP?
6. What is encapsulation? List out the advantages of encapsulation? What do you mean by DOM in JavaScript? How can you implement DOM object by using getElementById() method?

Group-'C'

Attempt any two questions (2X10=20)

8. Difference between client side and server side scripting language? Describe the different validation technique that can be used in order to make website secure with suitable example?
9. What is constructor? Why it is important in PHP? Explain with example.
10. Design an attractive user registration form using HTML and write mysql code to connect database?

*****Best of Luck*****

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Numerical Method (4th Sem)

Pre-Board Examination, Batch 2080, (Year 2082)

Group B

Attempt Any Five Question 5*6=30

1. Compute the root of equation $x^2 - 5x + 6 = 0$ using Newton Raphson Method

2. Estimate the value of $\ln(2.5)$ using Newton forward difference formula given the following data

x	1.0	2.0	3.0	4.0
ln(x)	0.0	0.6931	1.0986	1.3863

3. Locate the root of equation $x^2 + x - 2 = 0$ using Fixed point Method?

4. Evaluate $\int_0^1 \frac{1}{1+x^2} dx$ using simpsons 1/3 rule and simpson 3/8 rule

$$\int_0^1 \frac{1}{1+x^2} dx$$

5. Solve the following set of equation using gauss Jordan Method

$$3x + y + 2z = 15$$

$$2x + y + z = 7$$

$$x + 2y + 3z = 2$$

6. Use Runge kutta method to estimate $y(0.2)$ when $y'(x) = x + y$ with $y(0) = 0$ and $h = 0.2$.
From the following table, evaluate $y(40)$ using Gauss Forward formula

x	40	41	42	43	44
y	16.78	17.94	18.76	19.76	20.16

Group C

Attempt any Two Question 10*2=20

8. Find the inverse of following matrix using Gauss Elimination Method. A

$$A = \begin{bmatrix} 2 & -3 & 2 \\ 1 & -1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

9. Calculate the value of the following using Newton's data

x	1	2	3	4	5	6	7	8	9
y	1	2	3	4	5	6	7	8	9

10. Locate the root of equation $x^2 + x - 2 = 0$ using the fixed point method

10. Write an algorithm and program to compute the root of non-linear equation using Bisection Method

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Preboard Question(080 batch), DBMS(4th Sem)

2. What is Database management system.[1]

b) Given the following relations: [4]

Students(sid, sname, dept, year)

Courses(cid, cname, credits, dept)

Enroll(sid, cid, semester, grade)

Write relational algebra and SQL expressions for:

- Find names of all CSE department students enrolled in 'Database Systems' course
- Find students who have enrolled in all courses offered by their own department
- Find the course with maximum enrollment in Spring 2024 semester
- Find departments where average grade of students is above 8.5

3. What is query processing? Explain steps.

4. What is stored procedure. Explain with proper example [1+4]

5. What are the advantages of Normalization? Explain 4NF and 5NF.

6. What are the properties of Transaction. Explain the difference between mandatory and discretionary access control. [2+3]

7. Create the ER diagram for College Management System. [5]

8. Explain Full outer join and Inner join with ex. [1+4]

Originals: 2024/2025

9. What is Model. Explain Relational Database Model and Network Model [1+4]

Explain 3 tier architecture of dbms.[5]

10. Explain any 4 DDL commands [5]

What are the responsibilities of Database administrator [5]

11. Explain timestamp ordering protocol.[5]

What is concurrency control? Explain 2 phase lock protocol[5]

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Preboard, Batch[078], Operating System, 4th Sem.

Multiple Choice Question[1X10=10]

Attempt all the questions. [6x5=30]

1. What is operating system? Explain its functions with examples.
2. Explain sleeping barber problems with examples.
3. Explain five state process model.
4. Define distributed system with its characteristic.
5. What are the conditions of occurring Deadlock? Explain.
6. How do you secure your system? Explain threats in system.

Group C

Attempt all. [2x10=20]

7. Calculate average turnaround time and average waiting time of following process.

Process	Arrival time	Execution time
P1	0	5
P2	0	4
P3	1	3
P4	2	2
P5	2	1
P6	3	2
P7	4	2
P8	4	2

a) FIFO b) SJF c) SRTF d) Round Robin

Track the given process in table and check for deadlock. And also give the job sequence if deadlock can't occur using Banker's algorithm.

Process	Max. reqd.	Allocated	Available
P1	4	4	4
P2	7	5	3
P3	3	2	1
P4	9	0	4
P5	5	1	2
P6	4	1	3

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Pre Board Examination

BCA / Fourth Semester / CACS 253: Software Engineering

Candidates are required to answer the question in their own words as far as practicable.

Group A (6*5=30)

1. Define software engineering and explain its importance.
2. Explain different types of software with suitable examples.
3. What is feasibility study? Explain any three types of feasibility.
4. Explain the characteristics of a good Software Requirement Specification (SRS).
5. What is capability maturity model? Explain.
6. Explain black box testing and white box testing.
7. What is software maintenance? Explain its types.

Group B (2*10=20)

1. Describe the Waterfall model and Spiral model. Compare them with respect to flexibility, risk handling, and suitability.
2. Explain software testing in detail. Discuss different levels of testing and their objectives.
3. What is Model View Controller (MVC) architecture?
Explain the role of Model, View, and Controller.